

Foundation (Jalapeno)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
- (B) $ax + bx^2 + c$
- (C) $ax^2 - bx + c$
- (D) $ax^3 + bx + c$
- (E) $ax^2 + b$

Q2. Which of the following is a quadratic function in standard form?

- (A) $x^2 + 2x - 3$
- (B) $2x + 3$
- (C) $x^3 - 2x^2 + x - 1$
- (D) $x^2 - 4$
- (E) $3x^4 - 2x^2 + 1$

Q3. What is the value of a in the quadratic function $2x^2 + 3x - 1$?

- (A) 1
- (B) 2
- (C) 3
- (D) -1
- (E) -2

Q4. Which of the following quadratic functions has a negative leading coefficient?

- (A) $2x^2 + 3x - 1$
- (B) $-x^2 + 2x + 3$
- (C) $x^2 - 2x - 3$
- (D) $3x^2 + 2x - 1$
- (E) $-3x^2 + x - 2$

Q5. What is the x^2 coefficient in the quadratic function $-2x^2 + 3x - 1$?

- (A) 1
- (B) 2
- (C) -1
- (D) -2
- (E) 3

Q6. Which of the following is NOT a characteristic of a quadratic function in standard form?

- (A) The leading coefficient is 1
- (B) The degree of the polynomial is 2
- (C) The polynomial has a constant term
- (D) The polynomial has a linear term
- (E) The polynomial has a cubic term

Q7. What is the degree of the polynomial $3x^2 + 2x - 1$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q8. Which of the following quadratic functions has a positive leading coefficient?

- (A) $-x^2 + 2x + 3$
- (B) $2x^2 - 3x - 1$
- (C) $-3x^2 + x - 2$
- (D) $x^2 - 2x - 3$
- (E) $-2x^2 + x + 1$

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic function $f(x) = 2x^2 + 3x - 1$. Be sure to include the leading coefficient, degree, and any other relevant features.

Q10. Write a quadratic function in standard form with a leading coefficient of -1 , a linear coefficient of 2, and a constant term of -3 . Identify the degree and any other relevant features of the function.

Chef's Tips

- Tip 1: Make sure to check for any calculation errors
- Tip 2: Ensure students provide clear and concise explanations for FRQs
- Tip 3: Use the point system to grade FRQs